

Algorithm Analysis & Complexity Theory

Sample Exam: Np Completeness

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Difficulty Levels: Foundation, Conceptual, Application, Advanced

Question 1 [Difficulty: Conceptual]

Prove that if $P = NP$, then every NP-complete problem can be solved in polynomial time.

Solution:

Proof:

Definitions:

- P = class of problems solvable in polynomial time
- NP = class of problems verifiable in polynomial time
- NP-complete = hardest problems in NP (all NP problems reduce to them)

Assume: $P = NP$

Let L be any NP-complete problem

By definition of NP-complete:

- (1) $L \in NP$
- (2) For every problem $L' \in NP$, there exists a polynomial-time reduction from L' to L ($L' \leq_p L$)

Since $P = NP$:

- $L \in NP$ implies $L \in P$
- Therefore, L can be solved in polynomial time

For any other NP-complete problem L' :

- $L' \leq_p L$ (by definition of NP-complete)
- L can be solved in polynomial time

- A polynomial-time reduction composed with a polynomial-time algorithm gives a polynomial-time algorithm

- Therefore, $L' \in P$

Conclusion: If $P = NP$, then every NP-complete problem is in P ■

Note: This is why P vs NP is so important - solving one NP-complete problem in polynomial time would solve them all!